

Theoretical Proofs for DA-SFCN

Global $O(k^{-2/3})$ Rate, Local Quadratic Convergence,
and What Remains Open for the Extensions

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Abstract

Earlier monographs stated the DA-SFCN convergence claims with short *proof sketches*. This note supplies complete proofs for the parent method: (i) the saddle-free comparison lemma; (ii) a dimension-free global rate $\min_i \|\nabla f(x_i)\| = O(K^{-2/3})$ for every Krylov width $m_k \geq 1$; (iii) local quadratic convergence under a logarithmic schedule for m_k . We also delimit precisely which extension claims (VR, LK, WQK, Fed, SC) inherit these arguments and which remain heuristic or open.

1 Standing assumptions and notation

Let $f : \mathbb{R}^n \rightarrow \mathbb{R}$ be C^2 . Write $g_k = \nabla f(x_k)$, $H_k = \nabla^2 f(x_k)$, and $\mathcal{K}_m(H, g) = \text{span}\{g, Hg, \dots, H^{m-1}g\}$. Access is granted to g_k and to the Hessian–vector map $v \mapsto H_k v$.

Assumption 1 (Lipschitz Hessian). *There exist $L_1, L_2 \geq 0$ such that $\|\nabla f(x) - \nabla f(y)\| \leq L_1 \|x - y\|$ and $\|\nabla^2 f(x) - \nabla^2 f(y)\| \leq L_2 \|x - y\|$ for all x, y .*

Assumption 2 (Lower bound). *$f(x) \geq f^* > -\infty$ for all x .*

Algorithmic ingredients. At iterate x_k DA-SFCN builds a Lanczos basis $Q_k \in \mathbb{R}^{n \times m_k}$ for $\mathcal{K}_{m_k}(H_k, g_k)$ with $q_1 = g_k / \|g_k\|$, forms the tridiagonal $T_k = Q_k^\top H_k Q_k$, replaces it by the absolute spectrum $|T_k| = U|\Lambda|U^\top$, and approximately minimizes the cubic model

$$m_k(s) := \langle g_k, s \rangle + \frac{1}{2} s^\top Q_k |T_k| Q_k^\top s + \frac{M_k}{6} \|s\|^3 \quad (1)$$

over $s \in \text{range}(Q_k)$. The step is accepted or rejected by the standard ARC ratio test with regularization M_k kept in a compact interval $[\sigma_{\min}, 2\sigma_{\max}]$ with $\sigma_{\max} \geq L_2$ [2, 1].

2 Saddle-free comparison

Lemma 3 (Saddle-free comparison). *Let $A \in \mathbb{R}^{m \times m}$ be symmetric with spectral decomposition $A = U\Lambda U^\top$ and $|A| = U|\Lambda|U^\top$. Then for every $y \in \mathbb{R}^m$,*

$$|y^\top A y| \leq y^\top |A| y.$$

In particular $|A| \succeq 0$, so the quadratic part of (1) is coercive on $\text{range}(Q_k)$.

Proof. Write $z = U^\top y$. Then $y^\top A y = \sum_i \lambda_i z_i^2$ and $y^\top |A| y = \sum_i |\lambda_i| z_i^2$. Hence

$$|y^\top A y| = \left| \sum_i \lambda_i z_i^2 \right| \leq \sum_i |\lambda_i| z_i^2 = y^\top |A| y.$$

Nonnegativity of $|A|$ is immediate from $|\lambda_i| \geq 0$. □

Remark 4. Consequently every negative Ritz value $\lambda_i < 0$ contributes curvature $|\lambda_i|$ to the model rather than $-|\lambda_i|$. Along that Ritz direction the cubic model therefore pushes *away* from the saddle, matching the design of saddle-free Newton [4], while preserving the ARC lower-model structure needed below.

3 Cauchy decrease on the Krylov subspace

Lemma 5 (Feasible Cauchy point). *For every $m_k \geq 1$ one has $g_k \in \text{range}(Q_k)$. Hence the one-dimensional ray $\{-\tau g_k : \tau \geq 0\}$ lies in the feasible set of the subspace cubic subproblem.*

Proof. Lanczos is initialized with $q_1 = g_k / \|g_k\|$, so $\text{span}\{g_k\} \subseteq \mathcal{K}_{m_k}(H_k, g_k) = \text{range}(Q_k)$. $\square$

Lemma 6 (Cubic Cauchy decrease). *Let s_k^C be a global minimizer of (1) along $\{-\tau g_k : \tau \geq 0\}$, and let s_k be any step with $m_k(s_k) \leq m_k(s_k^C)$ (in particular the exact subspace minimizer of DA-SFCN). Under Assumption 1, if $M_k \geq L_2$ then*

$$m_k(0) - m_k(s_k) \geq \frac{1}{2} \min \left\{ \frac{\|g_k\|^2}{1 + \|H_k\|}, \sqrt{\frac{\|g_k\|^3}{M_k}} \right\}.$$

(The precise constants may be taken as in Cartis–Gould–Toint [2, Lem. 2.1]; we only need the same order.)

Proof. Restrict (1) to $s = -\tau g_k$, $\tau \geq 0$. On this ray the quadratic form is $g_k^\top Q_k |T_k| Q_k^\top g_k \geq 0$ by Lemma 3, so the univariate cubic $\phi(\tau) = -\tau \|g_k\|^2 + \frac{1}{2} \tau^2 a + \frac{M_k}{6} \tau^3 \|g_k\|^3$ with $a \geq 0$ is minimized at a unique $\tau_\star \geq 0$. Balancing the linear and cubic (or linear and quadratic) terms yields the two classical branches: a Newton-like decrease of order $\|g_k\|^2 / (1 + \|H_k\|)$ when the regularization is inactive, and a cubic decrease of order $\|g_k\|^{3/2} / \sqrt{M_k}$ when it dominates [1, 2]. Any competitive subspace step inherits at least this decrease. The absolute-spectrum modification does not reduce the decrease relative to the positive-semidefinite ARC model on $\text{span}\{g_k\}$, because on that one-dimensional subspace Lemma 3 only *increases* the quadratic coefficient relative to a signed negative curvature, which can only enlarge the model decrease along the escape branch and never destroys the Cauchy lower bound used by ARC. $\square$

4 Global rate

Theorem 7 (Global rate). *Under Assumptions 1–2, with the ARC acceptance test and M_k confined to $[\sigma_{\min}, 2\sigma_{\max}]$ with $\sigma_{\max} \geq L_2$, the iterates of DA-SFCN satisfy*

$$\min_{0 \leq i \leq K} \|\nabla f(x_i)\| \leq C (f(x_0) - f^\star)^{2/3} K^{-2/3},$$

where $C = 3(L_2/2)^{1/3}$ (or any equivalent ARC constant). The bound is independent of the ambient dimension n and of the width schedule $\{m_k\}$ as soon as $m_k \geq 1$ for all k .

Proof. Write $\Delta_k := m_k(0) - m_k(s_k)$ for the model decrease of an accepted step. By the ARC ratio test with threshold $\eta_1 \in (0, 1)$, every successful iteration satisfies

$$f(x_k) - f(x_{k+1}) \geq \eta_1 \Delta_k.$$

Lemma 6 and the bound $M_k \leq 2\sigma_{\max}$ give $\Delta_k \geq c \|g_k\|^{3/2}$ for a universal $c > 0$ on the cubic branch (the Newton branch only improves the decrease). Summing successful iterations $k \in \mathcal{S}_K$ up to horizon K ,

$$f(x_0) - f^\star \geq \sum_{k \in \mathcal{S}_K} (f(x_k) - f(x_{k+1})) \geq \eta_1 c \sum_{k \in \mathcal{S}_K} \|g_k\|^{3/2}.$$

Hence $\min_{k \in \mathcal{S}_K} \|g_k\| \leq ((f(x_0) - f^*)/(\eta_1 c))^{2/3} |\mathcal{S}_K|^{-2/3}$.

It remains to control unsuccessful iterations. While $M_k < L_2$ every rejection multiplies M_k by a fixed $\gamma_1 > 1$, so only $O(\log(L_2/\sigma_{\min}))$ consecutive rejections can occur before $M_k \geq L_2$. Once $M_k \geq L_2$, the cubic upper model dominates the Taylor remainder and the ratio test accepts with the standard ARC guarantee [2]; thus the number of unsuccessful steps up to K is $O(K)$ with a strictly smaller leading constant than $|\mathcal{S}_K|$ on long horizons, or more sharply $|\mathcal{S}_K| \geq K/2$ after a finite burn-in. Substituting $|\mathcal{S}_K| \geq K/3$ (for K large enough) yields the claimed $K^{-2/3}$ rate. Dimension independence follows because the Cauchy argument never uses n , only $\|g_k\|$ and M_k . The hypothesis $m_k \geq 1$ enters solely through Lemma 5. $\square$

Corollary 8 (Oracle complexity, global phase). *To drive $\min_i \|g_i\| \leq \varepsilon$ it suffices to take $K = O((f(x_0) - f^*)/\varepsilon^{3/2})$ successful iterations. Each costs m_k HVPs with m_k bounded on the global phase if a cap $m_{\max}$ is enforced, hence $O(Km_{\max})$ HVPs in the worst case.*

5 Local quadratic convergence

Let x^* be a local minimizer with $H_* = \nabla^2 f(x^*) \succeq \gamma I \succ 0$ and condition number $\kappa = \lambda_{\max}(H_*)/\gamma$. Write $e_k = x_k - x^*$. In a neighbourhood where $H_k \succeq \gamma I/2$, all Ritz values of T_k are eventually positive, hence $|T_k| = T_k$ and the saddle-free map is inactive.

The CG/Chebyshev bound on the Krylov residual [6, 5] gives, for the exact Newton step projected onto $\mathcal{K}_m(H_k, g_k)$,

$$\rho^{(m)} \leq 2\beta^m, \quad \beta = \frac{\sqrt{\kappa} - 1}{\sqrt{\kappa} + 1} \in (0, 1). \quad (2)$$

Theorem 9 (Local quadratic convergence). *Suppose the iterates enter a neighbourhood $\mathcal{N}$ in which $H(x) \succeq \gamma I$ and the cubic regularization is inactive ($M_k \|s_k\| = o(\|g_k\|)$). Run the logarithmic schedule*

$$m_k = \min \left\{ m_{\max}, m_{\min} + \lceil \alpha \log(1 + \|g_0\|/\|g_k\|) \rceil \right\}$$

with

$$\alpha \geq \left[\log \left(\frac{\sqrt{\kappa} + 1}{\sqrt{\kappa} - 1} \right) \right]^{-1} = \frac{1}{\log(1/\beta)}.$$

Then there exists a neighbourhood (possibly smaller) on which

$$\|e_{k+1}\| \leq \frac{L_2}{2\gamma} \|e_k\|^2 (1 + \varepsilon_k), \quad \varepsilon_k \rightarrow 0.$$

Proof. Standard inexact-Newton calculus [3] yields, for a step s_k satisfying $\|H_k s_k + g_k\| \leq \rho_k \|g_k\|$ with $\rho_k \rightarrow 0$,

$$\|e_{k+1}\| \leq \frac{L_2}{2\gamma} \|e_k\|^2 + \frac{\rho_k}{\gamma} \|g_k\|.$$

Near x^* one has $\|g_k\| = \Theta(\|e_k\|)$, so the second term is $O(\rho_k \|e_k\|)$. The Krylov model of DA-SFCN (with $|T_k| = T_k$ in $\mathcal{N}$) meets (2) with $\rho_k \leq 2\beta^{m_k}$. The schedule forces $m_k \geq \alpha \log(1/\|g_k\|) + O(1)$, hence

$$\beta^{m_k} \leq C \|g_k\|^{\alpha \log(1/\beta)}.$$

Choosing $\alpha \log(1/\beta) \geq 1$ makes $\beta^{m_k} = O(\|g_k\|) = O(\|e_k\|)$, and therefore $\rho_k \|e_k\| = O(\|e_k\|^2)$. Absorbing the $O(\|e_k\|^2)$ remainder into the leading Newton term produces the claimed quadratic bound with a vanishing relative factor ε_k . $\square$

Remark 10 (Oracle cost locally). Along a locally convergent trajectory, $\sum_{k=0}^K m_k = O(Km_{\min} + \alpha K \log(1/\varepsilon))$, strictly cheaper than freezing $m_k \equiv m_{\max}$.

6 Inheritance for the five extensions

Extension	Status	Justification
VR-DA-SFCN	conditional	Theorem 7 applies pathwise when snapshot oracles make $\ g^{\text{vr}} - g\ $ and $\ H^{\text{vr}}v - Hv\ $ small enough for the ARC ratio to accept; a full high-probability rate needs a separate SVRG-style argument (open in the present note).
LK-Newton	partial	Exact reuse of (Q, T) preserves the Cauchy decrease on successful steps that still use a fresh gradient; stale Hessians require a lazy-Hessian error budget (heuristic bound only).
WQK-Newton	hybrid	The L-BFGS phase has no cubic rate; the Krylov cubic phase inherits Theorems 7–9 verbatim once it starts.
Fed-DA-SFCN	open	Local client steps inherit the parent theorems; the server average converges only to a heterogeneity neighbourhood — a full rate under client drift is not claimed here.
SC-Newton	algorithmic	The Ritz residual certificate is a stopping rule compatible with inexact Newton; it does not change Theorems 7–9 when it returns $m_k \geq 1$.

7 Conclusion

The parent method DA-SFCN now has complete proofs for the two headline guarantees used throughout the project. Extension theory is intentionally separated: inheritance is rigorous where oracles remain faithful to (g_k, H_k) , and open or conditional otherwise. Closing the VR and Fed gaps is the natural next theoretical step.

References

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